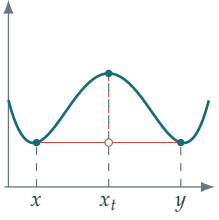
Panel 1

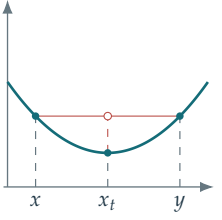
Nonconvex



$f(x_t) > \ell_t$

Panel 2

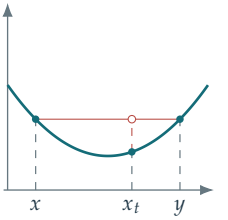
Convex



$f(x_t) \leq \ell_t$

Panel 3

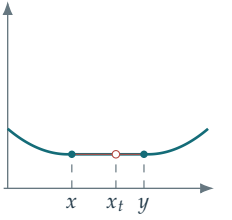
Strictly convex



$f(x_t) < \ell_t$

Panel 4

Convex, not strict



$f(x_t) = \ell_t$